

Worksheet 5E: Laws of Sines and Cosines — Solutions

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Law of Sines

1. $C = 80^\circ$, $b \approx 13.5$ cm, $c \approx 15.3$ cm
2. $B = 70^\circ$, $a \approx 8.9$ m, $b \approx 14.6$ m
3. $A = 80^\circ$, $a \approx 10.0$ cm, $c \approx 7.5$ cm
4. $C = 45^\circ$, $b \approx 11.6$ m, $c \approx 14.4$ m
5. $B \approx 53.1^\circ$ or $B \approx 126.9^\circ$; both valid (two solutions)
6. Two solutions
7. One solution
8. No solution ($a < b \sin A$)
9. (a) $\approx 0.000008^\circ$ (b) 8.6 light-years (simplified)
10. (a) $C = 45^\circ$ (b) $AC \approx 410$ m, $BC \approx 351$ m

Part B: Law of Cosines

11. $c \approx 7.7$
12. $a \approx 17.9$
13. $b \approx 8.7$
14. $c \approx 11.6$
15. $C \approx 95.7^\circ$
16. $C \approx 77.4^\circ$
17. $A \approx 26.4^\circ$
18. $A \approx 53.1^\circ$, $B \approx 59.5^\circ$, $C \approx 67.4^\circ$
19. ≈ 1.32 AU
20. ≈ 0.251 nm
21. ≈ 0.152 nm

22. ≈ 1116 km

23. ≈ 3150 km

Part C: Area

24. ≈ 30.8 square units

25. ≈ 140.6 square units

26. 30 square units

27. ≈ 26.8 square units

28. (a) 20 (b) 10 square units

29. $86,600 \text{ nm}^2$

Part D: Applications

30. (a) 64.6° , 64.6° , 50.8° (b) $\approx 1.63 \text{ m}^2$ (c) $\approx 522 \text{ m}^2$

31. (a) 60° , 60° , 60° (b) $\approx 0.00874 \text{ nm}^2$ (c) $\approx 0.123 \text{ nm}$

32. 8 AU; 7 AU

33. ≈ 1250 km

34. (a) ≈ 150 million km (b) $\approx 89.85^\circ$ (c) $\approx 0.15^\circ$
